

Physics-Guided, Identifiability-Aware Extraction of Stanford RRAM Compact-Model Parameters from Experimental TaO_x Devices

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Abstract—This paper presents a reproducible extraction flow for fitting the Stanford–PKU resistive random-access memory (RRAM) compact model to measured TaO_x DC switching data. The method is designed for a practical problem that is often hidden in compact-model calibration: a visually good I – V fit does not by itself prove that the fitted parameters are unique, physically meaningful, or identifiable from DC data. The proposed flow first extracts quality-controlled switching cycles from Keysight B1500 CSV exports and selects a measured medoid cycle as the nominal fitting target. It then classifies the dominant conduction mechanism in each resistance-state segment using ohmic, Poole–Frenkel, Schottky, and Fowler–Nordheim linearizations with Bayesian information criterion and a dynamic-permittivity admissibility test. The resulting regime map conditions the prior-extraction windows and the loss weights used during HSPICE-in-the-loop Bayesian optimization. After local refinement, the fitted parameter vector is accompanied by an operational identifiability report based on Fisher/Cramer–Rao analysis at the fitted optimum and fixed-nuisance one-dimensional loss profiles. A separate cluster-bootstrap stage over whole cycles propagates cycle-to-cycle variability into empirical confidence intervals. The result is not only a Stanford-model parameter set, but also a defensible statement of which parameters are supported by the measured TaO_x data and which are prior-dominated.

I. METHODOLOGY

A. Study Design

The objective of this work is to extract Stanford–PKU RRAM compact-model parameters from measured TaO_x devices in a way that is reproducible, physics-aware, and honest about parameter uncertainty. The central claim is therefore deliberately narrower than “we can fit an I – V curve.” The pipeline asks which model parameters are constrained by the available DC switching measurements, which parameters are only weakly constrained, and which parameters should not be interpreted as extracted material/device properties.

Three design choices follow from this objective.

First, conduction physics is treated as a front-end diagnostic rather than an after-the-fact explanation. TaO_x RRAM conduction can change with voltage window, resistance state, cycle, and deposition condition. The code therefore classifies each representative curve into ohmic, Poole–Frenkel (PF), Schottky, Fowler–Nordheim (FN), or unclassified windows before compact-model fitting. The classification is used only where the Stanford model can reasonably respond: ohmic/PF-like bulk regions receive full fitting weight, whereas Schottky/FN regions, which are not explicitly represented by the released Stanford current law, are down-weighted but not removed.

Second, a single measured cycle is used for the nominal point estimate. The representative curve is a medoid cycle selected from the experimental ensemble, not a synthetic pointwise median. This preserves the measured hysteresis ordering, switching kinks, and compliance-limited portions of the sweep. The cycle ensemble is kept separately and is used later for variability and bootstrap uncertainty.

Third, identifiability is reported as a primary output. The pipeline produces a best-fit parameter vector, but it also reports Fisher relative uncertainty, profile/loss curvature, and bootstrap spread. Parameters classified as non-identifiable are retained in the model vector for reproducibility, but they are not interpreted as independently extracted from DC data. A two-pass workflow can optionally freeze such parameters at their prior values and refit the remaining active subset.

Fig. 1 summarizes the complete implementation, and Algorithms 1 and 2 give the two core procedures.

B. Devices, Conditions, and Measurements

The experimental input consists of bipolar DC switching measurements from TaO_x RRAM devices organized into twelve sample/condition folders (S1–S12). The repository includes the deposition-condition table `Depos_Condition.csv`, which records the oxygen percentage and sputter power associated with each sample folder. The devices are treated in the pipeline as measured TaO_x cross-point RRAM cells, consistent with the TaO_x fabrication and switching studies reported in [3], [4].

The raw electrical measurements are Keysight B1500-style CSV exports. Each file may contain one or more analysis blocks, and each block contains voltage and current samples for a SET, RESET, mixed, or otherwise unlabeled sweep. The pipeline does not assume that every file is usable. It first parses all available measurement blocks, infers branch polarity, assigns a cycle number within each (`condition`, `device`) group, and rejects cycles that do not contain sufficient information for compact-model fitting.

The Stanford fits use the current compliance value configured for the final TaO-Fit runs, $I_{cc} = 5 \times 10^{-4}$ A. This value is a measurement-condition parameter rather than a fitted material parameter: the Stanford Verilog-A model does not include an instrument compliance circuit, so the simulated current is clamped to this magnitude after each transient simulation.

C. Stage 0: Raw Data Extraction and Quality Control

Raw data extraction is performed by `01_extract_clean_cycles.py`. The parser reads each CSV using tolerant text handling, detects B1500 analysis titles, extracts finite voltage-current pairs from `DataValue` rows, and falls back to treating the full file as one block if no title rows are available. The branch label is inferred from the block title first: `reset` gives RESET and `set` gives SET. If the title is uninformative, the code compares the positive and negative voltage extrema to infer SET, RESET, MIXED, or UNKNOWN.

Each parsed file becomes one cycle record. A cycle is retained only if all configured quality criteria are satisfied:

- at least 40 finite voltage-current points;
- voltage span of at least 1.0 V;
- peak absolute current between 1×10^{-9} A and 1 A;
- median absolute current above the numerical current floor 1×10^{-13} A;
- dynamic current range of at least 0.5 decades, computed from the 5th and 95th percentiles of $|I|$ above the current floor;
- both SET and RESET content, unless the command-line option disables this requirement.

These screens remove shorted devices, open devices, incomplete sweeps, noise-floor traces, and files with insufficient switching contrast. The outputs are (i) `01_extracted_points.csv`, a normalized point table; (ii) `01_cycle_summary.csv`, a per-cycle quality table; and (iii) `01_device_summary.csv`, a per-device summary containing the number of total cycles, number of good cycles, mean quality score, and good cycle fraction.

D. Stage 1: Representative Device, Programming Segment, and Medoid Cycle

The representative fitting target is produced by `02_select_device_representative_curve.py`. Unless the user manually specifies a condition and device, the selected device is the device with at least five good cycles and the best ranking by three ordered criteria: number of good cycles, mean quality score, and good-cycle fraction. If no device reaches the five-cycle threshold, all devices remain eligible and the same ranking is applied. This rule prevents the nominal fitting target from being chosen by visual preference alone.

For each good cycle and branch, the script then isolates the programming segment. Points are sorted by the instrument point index, and the voltage trace is split when the voltage direction changes. Candidate segments must contain at least 20 points and span at least 0.3 V. SET candidates are scored for positive polarity and positive-going voltage; RESET candidates are scored for negative polarity and negative-going voltage. The highest-scoring segment is retained as the programming segment for that cycle/branch.

The default representative mode is `medoid-cycle`. For each branch, all programming cycles are interpolated in $\log_{10} |I|$ onto a common voltage grid. The branch-typical

behavior is the median log-current at each grid voltage. The score of cycle c on branch b is

$$s_{c,b} = \text{median}_v |\log_{10} |I_{c,b}(v)| - \text{median}_{c'} \log_{10} |I_{c',b}(v)|, \quad (1)$$

computed only where enough cycles overlap. The script ranks cycles within each branch, converts the ranks to rank fractions, and chooses one real cycle that minimizes the mean branch rank fraction across the available SET and RESET branches. If no single cycle covers both branches adequately, the code falls back to the best measured cycle per branch. The output file keeps the downstream median-column schema, but in `medoid-cycle` mode these values are the measured values of the selected cycle. Consequently, the nominal point-estimate fit is anchored to an experimentally observed sweep, not to a synthetic average.

The separate script `02b_build_cycle_ensemble.py` builds a long-form ensemble table for the same selected device. This table contains every good cycle, tagged by `representative_cycle_id` and point index. The ensemble is not used to change the nominal point estimate in Stages 1–4; it is used in Stage 7 for leave-one-cycle-out and cluster-bootstrap analyses.

E. Stage 2: Conduction-Regime Classification

Stage 2 automates the conduction-mechanism analysis used in the experimental TaO_x study [3]. It produces a regime map that is later consumed by the prior extractor and the loss function.

1) *State segmentation*: The representative butterfly cycle is sorted by the sequence index and split by branch. The SET branch is divided at its maximum voltage into `SET_HRS_PRE` and `SET_LRS_POST`; the RESET branch is divided at its minimum voltage into `RESET_LRS_PRE` and `RESET_HRS_POST`. This sequence-based split is important because forward and return sweeps may occupy overlapping voltage ranges while corresponding to different resistance states.

2) *Candidate transport mechanisms*: Within each state segment, the code converts voltage and current to $V = |V_{\text{meas}}|$ and $I = |I_{\text{meas}}|$, removes points with $V < 0.05$ V or $I < 1 \times 10^{-13}$ A, and removes the apparent compliance plateau by discarding points with $I \geq 0.95 \max(I)$ in that cleaned segment. Sliding windows of length

$$w = \max(8, \lceil N/3 \rceil)$$

are evaluated with stride $\lfloor w/4 \rfloor$. On each window, four standard linearized regressions are tested:

$$\text{ohmic:} \quad \ln I = m \ln V + b, \quad 0.8 \leq m \leq 1.2, \quad (2)$$

$$\text{PF:} \quad \ln(I/V) = s_{\text{PF}} \sqrt{V} + b, \quad s_{\text{PF}} > 0, \quad (3)$$

$$\text{Schottky:} \quad \ln I = s_{\text{S}} \sqrt{V} + b, \quad s_{\text{S}} > 0, \quad (4)$$

$$\text{FN:} \quad \ln(I/V^2) = s_{\text{FN}}/V + b, \quad s_{\text{FN}} < 0. \quad (5)$$

PF and Schottky linearizations are nearly collinear on short voltage windows. The classifier therefore applies a physical admissibility test before comparing regression scores. For a

Algorithm 1: Conduction-regime classification

Input: Representative curve D ; optional cycle ensemble $\mathcal{E}$; t_{ox} , T , and admissible $\varepsilon_r^{\text{dyn}} \in [1, 60]$

Output: Regime map $\mathcal{R}$ and optional mechanism-stability table

- 1 sort D by the butterfly sequence index;
- 2 split SET at maximum voltage and RESET at minimum voltage to obtain SET_HRS_PRE, SET_LRS_POST, RESET_LRS_PRE, and RESET_HRS_POST;
- 3 **foreach** state segment S **do**
- 4 convert to $V = |V|$, $I = |I|$; remove low-voltage, current-floor, and compliance-plateau points;
- 5 set $w = \max(8, \lceil |S|/3 \rceil)$ and slide by $\lfloor w/4 \rfloor$;
- 6 **foreach** window W **do**
- 7 regress Eqs. (2)–(5);
- 8 reject candidates that fail slope constraints or the $\varepsilon_r^{\text{dyn}}$ check;
- 9 assign the admitted mechanism with minimum BIC, or unclassified;
- 10 **end**
- 11 merge adjacent same-label windows and refit each merged span;
- 12 set stanford_valid=1 for ohmic/PF and 0 otherwise;
- 13 **end**
- 14 **if** $\mathcal{E}$ is provided **then**
- 15 classify each cycle by state segment and report mechanism fractions;
- 16 **end**

fitted square-root-voltage slope s , the dynamic relative permittivity implied by the PF or Schottky expression is

$$\varepsilon_r^{\text{dyn}} = \frac{q^3}{\pi \varepsilon_0 t_{\text{ox}} (skT)^2} \begin{cases} 1, & \text{PF,} \\ 1/4, & \text{Schottky.} \end{cases} \quad (6)$$

The candidate is admitted only if $\varepsilon_r^{\text{dyn}} \in [1, 60]$ under the configured $t_{\text{ox}} = 5 \text{ nm}$ and $T = 300 \text{ K}$. This broad range is intentionally conservative; it prevents an arbitrary square-root-voltage trend from being labeled PF or Schottky while still allowing realistic TaO_x dielectric variation.

For all admitted candidates, model selection uses the Bayesian information criterion

$$\text{BIC} = n \ln \left(\frac{\text{RSS}}{n} \right) + 2 \ln n, \quad (7)$$

where each linearized fit has two fitted coefficients. The lowest-BIC admissible mechanism wins. Windows for which no candidate passes the sign/range and permittivity checks are labeled *unclassified*. Consecutive windows with the same label are merged, and the winning linearization is refit over the merged voltage span to report the final slope, intercept, R^2 , BIC, dynamic permittivity, and point count.

3) *Regime-map outputs:* The regime map contains the columns *branch*, *state*, *v_lo*, *v_hi*, *mechanism*, *slope*, *intercept*, *r2*, *bic*, *eps_r_dyn*, *n_points*, and *stanford_valid*. The last flag is true for ohmic and PF windows and false for Schottky, FN, and unclassified regions. When the per-cycle ensemble is provided, each cycle is also classified at the state-segment level; the fraction of cycles assigned to each mechanism is reported as a cycle-to-cycle mechanism-stability metric.

F. Stanford Compact Model and HSPICE Forward Map

The fitted device model is the Stanford-PKU RRAM Verilog-A model `rram_v_1_0_0` [1], [2], simulated with

HSPICE through generated transient netlists. The model state variable is the conductive-filament gap g . The terminal current is

$$I_{\text{tb}} = I_0 \exp \left(-\frac{g}{g_0} \right) \sinh \left(\frac{V_{\text{tb}}}{V_0} \right), \quad (8)$$

and the deterministic gap rate implemented in the Verilog-A source is

$$\frac{dg}{dt} = -\nu_0 \exp \left(-\frac{qE_a}{kT_{\text{cur}}} \right) \sinh \left(\frac{\gamma a_0}{t_{\text{ox}}} \frac{qV_{\text{tb}}}{kT_{\text{cur}}} \right), \quad (9)$$

with

$$T_{\text{cur}} = T_{\text{ini}} + |V_{\text{tb}} I_{\text{tb}} R_{\text{th}}|. \quad (10)$$

The gap is integrated from `gap_ini` and clamped to `[gap_min, gap_max]`. The released HSPICE model computes the field-enhancement factor as

$$\gamma = \gamma_{\text{ini}} - \beta \left(\frac{g}{1 \text{ nm}} \right)^3, \quad \gamma_{\text{ini}} = \begin{cases} \gamma_0, & V_{\text{tb}} \geq 0, \\ 16, & V_{\text{tb}} < 0, \end{cases} \quad (11)$$

and sets $\gamma = 0$ when $\gamma |V_{\text{tb}}|/t_{\text{ox}} < F_{\text{min}}$. This negative-bias override is an important implementation detail: although reset-side `gamma0_res` is retained in the parameter/prior table, the default optimization does not actively fit it because the Verilog-A branch overrides γ_0 under negative bias.

The HSPICE deck places a series resistor R_s between the voltage source and the RRAM device. This resistor accounts for probe/parasitic resistance and softens the LRS branch; it is not a substitute for the SMU compliance. For each objective evaluation the code runs two separate transients: one SET transient, initialized in HRS, and one RESET transient, initialized in LRS. Polarity-split parameters allow the compact model to fit asymmetric measured SET and RESET behavior while keeping the scale/parasitic parameters shared.

The measured voltage samples are replayed as a piecewise-linear source. The time increment between adjacent samples is

$$\Delta t_i = \max \left(\frac{|V_i - V_{i-1}|}{1 \times 10^5 \text{ V s}^{-1}}, 1 \times 10^{-12} \text{ s} \right),$$

and the transient step is configured as $1 \times 10^{-6} \text{ s}$. This is a quasi-static mapping chosen for tractable HSPICE simulation; the fitted ν_0 and E_a absorb the effective sweep-time scale. The simulated source current is parsed from the HSPICE `.lis` or `.mt0` table, converted from source-current sign convention to the project device-current convention, interpolated back to the measured time stamps, and then clamped to $|I| \leq I_{\text{cc}}$.

G. Stage 3: Regime-Conditioned Priors

Priors are extracted in `stagel_extract_priors.m`. Each parameter receives a mean μ , standard deviation σ , and hard lower/upper bounds. The prior vector contains 21 entries: seven shared parameters, twelve polarity-split parameters, and two branch-specific initial gaps. The default final configuration uses $T = 300 \text{ K}$, $t_{\text{ox}} = 5 \text{ nm}$, $a_0 = 0.25 \text{ nm}$, and $\varepsilon_r = 25$.

When a regime map is supplied, the prior windows are selected from the classified physics. This is the main distinction from a fixed-window compact-model fit.

1) *LRS sinh prior for I_0 and V_0* : The LRS current scale is estimated from the low-voltage ohmic window of `SET_LRS_POST`. If no regime map is available, the legacy fallback window is $0.02 \leq V \leq 0.6$ V. The fit uses the SET return half when possible and excludes the compliance plateau. The regression model is

$$I(V) = A \sinh\left(\frac{V}{V_0}\right), \quad (12)$$

solved in log-parameter space with Nelder–Mead. Since the LRS gap is seeded near $g_{\min}$, the current-scale prior is

$$\mu_{I_0} = A \exp\left(\frac{g_{\min}}{g_{0,\text{prior}}}\right), \quad (13)$$

with uncertainty propagated from the residual scale and the g_0 prior. The implemented lower bound on the I_0 prior uncertainty is 25% of $|\mu_{I_0}|$.

2) *PF priors for γ_0 and E_a* : The HRS PF window of `SET_HRS_PRE` is used for the SET-side `gamma0_set/Ea_set` priors, and the HRS PF window of `RESET_HRS_POST` is used for the reset-side `gamma0_res/Ea_res` priors. For a PF regression

$$\ln(I/V) = \kappa\sqrt{V} + b,$$

the implemented mapping is

$$\mu_{\gamma_0} = 2\kappa\sqrt{\pi\epsilon t_{\text{ox}}}\frac{kT}{q}\frac{t_{\text{ox}}}{a_0^2}, \quad (14)$$

$$\mu_{E_a} = -b\frac{kT}{q}, \quad (15)$$

followed by clipping to the hard bounds in Table I. If a regime map is present but no PF window exists for a branch, the code does not force a PF fit onto non-PF data. Instead, it keeps the literature-centered defaults $\gamma_0 = 16.5$ and $E_a = 0.6$ eV, inflates their standard deviations to 12 and 0.4, respectively, and records the missing PF prior in the `pf_available` flag.

3) *Remaining priors and active set*: The series-resistance mean is estimated from the measured LRS resistance when available; otherwise the fallback is sized from $5\text{ V}/I_{\text{cc}}$ and then clamped to $[10, 10^5]\Omega$ before optimization bounds are applied. The remaining priors are literature- or structure-centered values with deliberately broad bounds.

By default, the optimizer varies 16 parameters:

$$\begin{aligned} &I_0, g_0, R_s, \gamma_{0,\text{set}}, \\ &V_{0,\text{set}}, V_{0,\text{res}}, E_{a,\text{set}}, E_{a,\text{res}}, F_{\min,\text{set}}, F_{\min,\text{res}}, \\ &\nu_{0,\text{set}}, \nu_{0,\text{res}}, g_{\max,\text{set}}, g_{\max,\text{res}}, g_{\text{ini},\text{set}}, g_{\text{ini},\text{res}}. \end{aligned}$$

The other prior-table parameters, including β , R_{th} , $g_{\min}$, t_{ox} , and `gamma0_res`, remain fixed at their prior means unless the user overrides `active_params` in the YAML configuration.

H. Stage 4: Mechanism-Aware Objective Function

The optimizer minimizes a variance-scaled log-current loss plus switching threshold and prior penalties. For measured point i ,

$$r_i(\theta) = \frac{\log_{10}|I_i^{\text{sim}}(\theta)| - \log_{10}|I_i^{\text{meas}}|}{\sigma_i^{\log I}}. \quad (16)$$

The noise scale is computed from the current interquartile range when such an ensemble-derived spread is present:

$$\sigma_i^{\log I} = \frac{|q_{75,i} - q_{25,i}|}{1.35 |I_i^{\text{med}}| \ln 10}. \quad (17)$$

Non-finite or non-positive values are replaced by 0.2 decades, and all values are floored at 0.05 decades. This detail matters for the medoid point estimate: because the medoid CSV contains a single measured cycle in the median/q25/q75 schema, $q_{75} = q_{25}$ and the point-estimate fit uses the 0.2-decade fallback. In the bootstrap stage, the resampled ensemble is aggregated by branch and point index, so Eq. (17) reflects cycle-to-cycle spread.

If mechanism weighting is enabled and a regime map is available, each point is assigned a weight

$$w_i = \begin{cases} 1, & \text{ohmic, PF, or no covering classified window,} \\ w_{\text{nb}}, & \text{covered only by Schottky or FN windows,} \end{cases} \quad (18)$$

with $w_{\text{nb}} = 0.3$ in the final configuration. The SET and RESET current losses are weighted mean-squared residuals,

$$L_b = \frac{\sum_{i \in b} w_i r_i^2}{\sum_{i \in b} w_i}, \quad b \in \{\text{SET}, \text{RESET}\}. \quad (19)$$

The state assignment for a point is based on its sequence position relative to the branch voltage extremum, not solely on voltage magnitude, so forward and return segments remain distinct.

The total objective is

$$L(\theta) = L_{\text{SET}} + L_{\text{RESET}} + 0.5L_V + 0.1L_{\text{prior}}. \quad (20)$$

Here L_V penalizes mismatch between measured and simulated switching thresholds, where the threshold is the voltage of maximum $d \log_{10}|I|/dV$ for SET and minimum $d \log_{10}|I|/dV$ for RESET. The voltage scale is the median measured voltage step, floored at 0.01 V. The prior term is

$$L_{\text{prior}} = \sum_j \left(\frac{\theta_j - \mu_j}{\sigma_j} \right)^2$$

over finite prior standard deviations. In the `plain` ablation mode, the prior standard deviations are set to infinity, removing this prior penalty and expanding the search to the hard physical bounds.

I. Stage 5: Global Optimization and Local Refinement

Global search is performed by MATLAB `bayesopt` in `stage3_bayesopt_driver.m`. The final configuration uses 48 quasi-random seed evaluations and 160 adaptive evaluations with the expected-improvement-plus acquisition function, deterministic objective setting, no parallelism, and random seed 0. Parameters with strong scale behavior are optimized with logarithmic transforms where configured. The wide-search parameters I_0 , g_0 , R_s , $\nu_{0,\text{set}}$, $\nu_{0,\text{res}}$, $F_{\min,\text{set}}$, and $F_{\min,\text{res}}$ use their full hard bounds. Other active parameters use $[\mu_j - 3\sigma_j, \mu_j + 3\sigma_j]$ clipped to the hard bounds, unless the plain-prior ablation expands them to the full bounds.

TABLE I
PRIOR TABLE IMPLEMENTED BY `STAGE1_EXTRACT_PRIORS.M`. VALUES SHOWN IN NANOMETERS OR V/NM ARE STORED IN SI UNITS IN THE CODE.

Parameter	Source	Mean	Std. dev.	Bounds	Unit
I_0	LRS sinh	Eq. (13)	$\geq 0.25 \mu $	$[10^{-8}, 9.9 \times 10^{-3}]$	A
g_0	config/lit.	0.275	0.05	$[0.15, 0.50]$	nm
β	config/lit.	1.25	0.5	$[0.1, 5]$	–
R_{th}	config/lit.	2.1×10^3	10^3	$[1, 10^6]$	K/W
g_{min}	structure	0.10	0.05	$[0.05, 1.0]$	nm
t_{ox}	config	5.0	0.5	$[2, 10]$	nm
R_s	LRS/fallback	R_{LRS} or $5/I_{cc}$	0.5μ	$[10, 10^6]$	Ω
$V_{0,\{set,res\}}$	LRS sinh	Eq. (12)	fit scale	$[0.05, 2.0]$	V
$\gamma_{0,\{set,res\}}$	PF or default	Eq. (14) or 16.5	0.4μ or 12	$[4, 24]$	–
$E_{a,\{set,res\}}$	PF or default	Eq. (15) or 0.6	0.25 or 0.4	$[0.1, 0.99]$	eV
$F_{min,\{set,res\}}$	config/lit.	1.4	0.42	$[0.5, 2.99]$	V/nm
$\nu_{0,\{set,res\}}$	config/lit.	10	5	$[0.1, 19.9]$	m/s
$g_{max,\{set,res\}}$	structure	1.7	0.3	$[1, 10]$	nm
$g_{ini,set}$	HRS seed	1.4	0.3	$[0.5, 2.5]$	nm
$g_{ini,res}$	LRS seed	0.10	0.05	$[0.05, 0.5]$	nm

The Bayesian-optimization result is refined by `stage4_local_refine.m` using Nelder–Mead `fminsearch` in log-parameter coordinates. The default local-refinement budget is 100 function evaluations. Any proposal outside the hard bounds receives a large penalty, preserving positivity and physical bounds without requiring constrained-simplex machinery. A nominal point-estimate run therefore costs $2(48 + 160 + 100) = 616$ HSPICE branch transients, because each objective evaluation requires one SET and one RESET simulation.

J. Stage 6: Operational Identifiability

Identifiability is evaluated after optimization at the fitted optimum $\hat{\theta}$. This ordering is intentional: the Fisher information and the local sensitivity directions are properties of the fitted solution, not of the prior mean. The code writes a per-condition `identifiability_report.csv` containing the parameter value, active flag, Fisher relative uncertainty, Fisher class, profile verdict, and PF-prior availability.

1) *Fisher/Cramer–Rao analysis*: The finite-difference sensitivity matrix is computed for every parameter in the prior vector, not only for the active subset. For parameter θ_j , the code perturbs the fitted optimum with a relative step of 10^{-3} , clips the lower perturbation at the lower bound, and forms the log-parameter sensitivity

$$S_{ij} = \frac{\log_{10} |I_i(\theta_j^+)| - \log_{10} |I_i(\theta_j^-)|}{\ln \theta_j^+ - \ln \theta_j^-}. \quad (21)$$

After variance scaling, the Fisher matrix is

$$F = \tilde{S}^T \tilde{S}. \quad (22)$$

Because compact-model parameterizations are often sloppy, F may be rank-deficient. The effective standard deviation is therefore computed from the pseudo-inverse,

$$\sigma_j^{\text{eff}} = \sqrt{[F^+]_{jj}},$$

and the relative uncertainty is

$$\rho_j = \frac{\sigma_j^{\text{eff}}}{|\hat{\theta}_j|}.$$

The implemented classification is:

$$\begin{cases} \rho_j < 0.5, & \text{identifiable,} \\ 0.5 \leq \rho_j < 2, & \text{weakly identifiable,} \\ \rho_j \geq 2, & \text{non-identifiable.} \end{cases}$$

The eigenspectrum and eigenvectors of F are also archived to expose near-null parameter combinations that are not visible from scalar uncertainty alone.

2) *Fixed-nuisance loss profiles*: Fisher analysis is local and linearized, so the final configuration also runs a one-dimensional loss-profile stage for active parameters. For each active parameter, `stage4b_profile_likelihood.m` evaluates the objective at seven grid points across

$$[\max(l_j, \hat{\theta}_j - 2\sigma_j), \min(u_j, \hat{\theta}_j + 2\sigma_j)]$$

with all other parameters held fixed at the fitted optimum. This is a fixed-nuisance profile scan rather than a full nuisance-parameter reoptimization, chosen to make the check feasible with HSPICE-in-the-loop simulation. Let Δ_{lo} and Δ_{hi} be the loss increases at the two scan edges relative to the optimum, and let $\Delta = \min(\Delta_{lo}, \Delta_{hi})$. The implemented verdict is `flat_non_identifiable` for $\Delta < 1$, `weakly_identifiable` for $1 \leq \Delta < 4$, and `identifiable` for $\Delta \geq 4$.

3) *Interpretation policy*: The reported fitted vector remains the actual vector used in simulation, even when some entries are weak or non-identifiable. The manuscript interpretation is stricter: identifiable parameters may be discussed as data-supported extractions; weakly identifiable parameters should be discussed with caution; non-identifiable parameters should be described as prior-dominated under the available DC data. If desired, the two-pass `auto_active` workflow uses a previous `identifiability_report.csv` to freeze non-identifiable parameters and refit the remaining active subset.

K. Stage 7: Validation and Cycle-to-Cycle Uncertainty

Validation is performed in `stage5_validate.m`. For the final fitted vector, the code simulates the SET and RESET branches, computes unweighted log-current residuals in

decades, and reports branch-wise RMSE together with the Durbin–Watson statistic:

$$DW = \frac{\sum_{i=2}^n (e_i - e_{i-1})^2}{\sum_{i=1}^n e_i^2}. \quad (23)$$

The RMSE measures magnitude of fit error; the Durbin–Watson statistic indicates whether residuals are structured along the sweep. Strong residual autocorrelation is interpreted as systematic compact-model mismatch rather than random measurement noise.

Cross-cycle uncertainty is computed only when the per-cycle ensemble is provided. The ensemble must contain at least three cycles and a point index. Leave-one-cycle-out (LOCO) validation is optional. In each LOCO iteration, the code aggregates all but one cycle by branch and point index, fits the training aggregate with a reduced Bayesian-optimization budget, and evaluates the held-out aggregate.

The main cycle-to-cycle uncertainty result is the cluster bootstrap implemented by `06_run_variability.py`. For each bootstrap draw, whole cycles are sampled with replacement from the selected device. The sampled cycles are collapsed to one median curve by (`branch`, `point_index`), preserving the hysteresis sequence. That curve is refit with a reduced budget of 8 seed evaluations plus 24 adaptive evaluations. Across $B = 120$ default resamples, the 2.5th and 97.5th percentiles of each parameter form empirical 95% confidence intervals. Because resampling is done at the cycle level, the intervals reflect cycle-to-cycle variability of the representative device; they do not claim device-to-device or wafer-level population intervals.

L. Ablation Design

The ablation harness `05_run_ablation.py` fits each condition under four variants using the same compact model and comparable evaluation budgets:

- 1) **physics_regime_bo**: the proposed method, with regime-conditioned priors, mechanism-aware weighting, and Bayesian optimization;
- 2) **physics_bo**: physics priors from the legacy fixed windows and no regime-map weighting;
- 3) **plain_bo**: no informative prior penalty (`prior_mode=plain`) and Bayesian optimization over the hard physical bounds;
- 4) **physics_nobo**: regime-conditioned prior means only, with optimization skipped.

The comparison isolates the contribution of the regime classifier, the physics priors, and the optimizer. The ablation summary reports the unweighted validation RMSE for SET and RESET so that methods using mechanism-aware loss are not favored by their training weights.

M. Implementation Traceability and Reproducibility

The end-to-end publication workflow is orchestrated by `run_full_pipeline.sh`. It calls `07_run_final.py` for per-condition point estimates, regime classification, Fisher identifiability, and profile scans; `04_publication_summary.py` for aggregated

Algorithm 2: Physics-guided, identifiability-aware Stanford parameter extraction

Input: Representative curve D ; optional cycle ensemble $\mathcal{E}$; regime map $\mathcal{R}$; YAML configuration
Output: Fitted parameter vector, identifiability report, validation metrics, and optional bootstrap confidence intervals

- 1 parse D into SET and RESET branches and extract switching/read features;
- 2 derive priors from the LRS ohmic window and HRS PF windows of $\mathcal{R}$; if no PF window exists for a branch, keep default γ_0/E_a priors with inflated uncertainty;
- 3 assemble the 21-entry prior table and choose the active set from the YAML configuration or the default 16-parameter set;
- 4 minimize Eq. (20) by Bayesian optimization (48 + 160 evaluations in final runs), using two HSPICE transients per objective call;
- 5 polish the best Bayesian-optimization point by Nelder–Mead in log-parameter coordinates;
- 6 compute the Fisher/Cramer–Rao report at the fitted optimum and classify each parameter by relative uncertainty;
- 7 **if** `run_profile=1` **then**
- 8 scan each active parameter over a seven-point fixed-nuisance loss profile and write the profile verdict;
- 9 **end**
- 10 simulate the final vector and write branch-wise RMSE and Durbin–Watson statistics;
- 11 **if** $\mathcal{E}$ is provided and bootstrap is enabled **then**
- 12 resample whole cycles, aggregate by branch/point index, refit with the reduced budget, and report empirical 95% confidence intervals;
- 13 **end**

parameter/fit-quality figures; `05_run_ablation.py` for the ablation; and `06_run_variability.py` for bootstrap confidence intervals and mechanism-stability maps.

All final fitting constants are centralized in `stanford_fit/config/final_config.yaml`. The final point-estimate configuration uses `run_fisher=1`, `run_profile=1`, `mechanism_weighting=1`, `nonbulk_weight=0.3`, `bo_n_init=48`, and `bo_n_iter=160`. Bootstrap and LOCO are kept off in the point-estimate configuration because they are separate, much heavier reporting stages driven by `06_run_variability.py`. Every fit archives its priors, Bayesian-optimization result, refined parameter vector, Fisher matrix/report, profile scans, validation residuals, and fit-quality plot.

N. Threats to Validity and Reviewer-Facing Limitations

The method makes the following limitations explicit. First, the released Stanford current law in Eq. (8) does not include separate Schottky or Fowler–Nordheim branches. The proposed weighting therefore reduces the influence of such windows on the fitted Stanford parameters, but it does not make the Stanford model physically complete in those regimes. Residual structure in Schottky/FN-dominated windows should be reported as model mismatch, not hidden.

Second, PF/Schottky discrimination depends on the admissible $\varepsilon_r^{\text{dyn}}$ bracket. The bracket $[1, 60]$ is deliberately broad, and the fitted $\varepsilon_r^{\text{dyn}}$ is written to the regime map so assignments can be audited or sensitivity-tested.

Third, the transient replay is a quasi-static numerical mapping, not the literal measurement sweep rate. Parameters such

Data preparation and transport classification (Python)

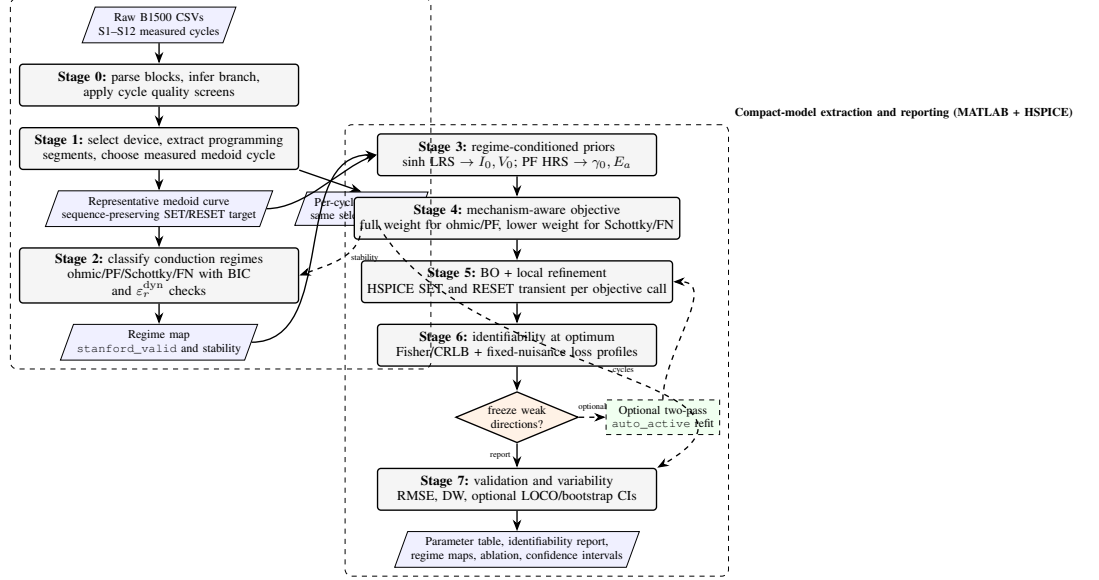


Fig. 1. End-to-end extraction workflow implemented in the repository. The regime map is the connection between the experimental conduction analysis and the compact-model fit: it selects prior windows, sets mechanism-aware weights, and helps interpret which parameters can be extracted from the data.

as ν_0 and E_a can therefore absorb effective time-scale assumptions. They should be compared within this protocol and across similarly processed conditions, not treated as protocol-independent kinetic constants.

Fourth, the reset-side γ_{gamma0} path is affected by the Verilog-A negative-bias override in Eq. (11). The branch-split simulation allows the reset branch to fit through other reset-side parameters, but it does not remove this structural asymmetry of the released model.

Fifth, the profile-likelihood stage is a fixed-nuisance one-dimensional scan. It is useful practical evidence of flat directions, but it is not a full profile with nuisance-parameter reoptimization. Higher-dimensional sloppy manifolds are therefore assessed through the Fisher eigenspectrum and the bootstrap rather than claimed to be fully resolved by the one-dimensional profiles.

Finally, bootstrap confidence intervals are cluster intervals over cycles of the selected representative device. They quantify cycle-to-cycle variability for that device and fitting protocol. They do not replace a separate device-to-device variability study.

REFERENCES

- [1] X. Guan, S. Yu, and H.-S. P. Wong, “A SPICE compact model of metal oxide resistive switching memory with variations,” *IEEE Electron Device Lett.*, vol. 33, no. 10, pp. 1405–1407, Oct. 2012.
- [2] Z. Jiang *et al.*, “A compact model for metal–oxide resistive random access memory with experiment verification,” *IEEE Trans. Electron Devices*, vol. 63, no. 5, pp. 1884–1892, May 2016.
- [3] M. T. R. Chowdhury, A. Moazzeni, and G. Tutuncuoglu, “Atomic-scale insights into the switching mechanisms of RRAM devices,” in *Proc. IEEE 68th Int. Midwest Symp. Circuits Syst. (MWSCAS)*, Lansing, MI, USA, 2025, pp. 518–522.
- [4] A. Moazzeni, M. T. R. Chowdhury, and G. Tutuncuoglu, “The impact of operation conditions on potentiation, depression and endurance dynamics of tantalum oxide RRAMs,” in *Proc. IEEE Int. Integr. Rel. Workshop (IIRW)*, 2023.

- [5] F.-C. Chiu, “A review on conduction mechanisms in dielectric films,” *Adv. Mater. Sci. Eng.*, vol. 2014, Art. no. 578168, 2014.

- [6] A. Raue *et al.*, “Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood,” *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009.